

Jonathan Santosa

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EDUCATION

Santa Clara University | *B.S. Computer Science and Engineering* (GPA: 3.84) Expected Graduation June 2025

- Relevant Coursework: Artificial Intelligence, Object Oriented Programming, Theory of Algorithms, Software Engineering, Advanced Programming, Data Structures, Web Development, Embedded System, Logic Design.

WORK EXPERIENCE

Cadent | **Backend Engineer Intern** *San Jose, CA* Jun 2024 - Aug 2024

- Developed an advanced AI-powered search engine for internal engineering teams using OpenAI API and ElasticSearch, enhancing search capabilities with sorting, filtering, and aggregation.
- Collaborated with front-end teams to integrate ElasticSearch with a Angular JS-based UI, ensuring seamless search functionalities.
- Developed and integrated multi-step interactive prompts using OpenAI models (GPT-3.5-turbo and GPT-4.0), leveraging prompt engineering to enhance user experience.
- Utilized Swagger for comprehensive API documentation and Apache MVN for efficient software deployment and management.

Santa Clara University Imaginarium Lab | **Research Software Engineer** *Santa Clara, CA* Jun 2022 - Present

- Create an application that enables users to customize their virtual characters and navigate through a virtual environment.
- Leverage Quest Pro 3 hardware and integrate the ReadyPlayerMe API.
- Allow users to change various body dimensions such as chest, head, arms, legs, waist, and hips, enhancing the overall immersive experience. Build with the intention of aiding users dealing with body dysmorphia.

FlyHigh Group, Inc. | **Unity Developer** *Santa Clara, CA* Apr 2022 - Dec 2023

- Developed a Unity-based application connecting companies and job seekers in an interactive 3D space, integrating Google's Text-To-Speech API to guide users through various locations.
- Built features allowing companies to create guilds, host meetings, customize spaces, and hold webinars, while enabling job seekers to personalize profiles and explore company environments.
- Enhanced multiplayer interactivity by implementing text and voice chat functionalities using PhotonView and PhotonVoice, improving real-time communication.

Santa Clara University Leavey Business School | **Research Software Engineer** *Santa Clara, CA* Jun 2022 - Nov 2022

- Collaborated with San Jose State University Lucas College of Business to develop a VR build using HP Omnicept, enhancing user experience with advanced sensory feedback.
- Created hyper-realistic and low-polygon avatars, along with character animations and special effects, to deliver an immersive VR environment.
- Developed an interactive space in Blender for users to express emotions, integrating it into VR and AR experiences using Unity plug-ins.

PUBLICATION

- Nguyen, T., Lee, E., Santosa, J., Galstyan, H., Dang, E., Khadadeh, M., & Jeong, D. (2023, Summer). Restless VR: Simulating Escapism among Individuals with Social Anxiety. 73rd Annual International Communication Association Conference. Toronto: International Communication Association.

ACCOMPLISHMENTS

Won First Place 2022 Hack for Humanity - OCEAN POLLUTION VIRTUAL MUSEUM Feb 2022

- Built a virtual museum (in 24 hours) using Unity Engine and Blender to spread awareness about ocean pollution.

Won First Place in Santa Clara University Winter Challenge - SUBJUGATED EXTRACTION Dec 2021

TECHNICAL SKILLS

Programming: C#, Python, Java, C++, Javascript, CSS, SQL, React, Docker, ChatGPT API, Google Text-speech API, Angular, Swagger, OpenAI API, Apache MVN

Game Engines: Unity, Unreal Engine 4 & 5, Blender, Maya, MIT Scratch | **Version Control:** GitHub, PlasticSCM

Content Creations: IMovie, Final Cut Pro, Spatial.io | **API:** ChatGPT, Google Text-speech, ReadyPlayerMe